

HAAD Practitioner Workpapers

Operational Forms

Version 0.5

Kevin V. Watson

August 25, 2026

Copyright © 2026 Kevin V. Watson. Licensed under CC BY-NC 4.0.

Document information

Author: Kevin V. Watson

Version 0.5 DOI: [10.5281/zenodo.22099968](https://doi.org/10.5281/zenodo.22099968)

Prior deposited version: [Version 0.4, DOI 10.5281/zenodo.21683276](https://doi.org/10.5281/zenodo.21683276)

Canonical source: *Human-Agent Attribution Discontinuity: A Forensic and Assurance Methodology*, Version 0.5

Status: Operational companion. The canonical methodology controls if this extraction and the full text differ.

Copyright and licence: Copyright © 2026 Kevin V. Watson. Licensed under the [Creative Commons Attribution-NonCommercial 4.0 International Licence](https://creativecommons.org/licenses/by-nc/4.0/). Adaptations must identify their changes and must not be represented as official HAAD materials or as endorsed by the author.

Record the methodology version, application level, examiner, reviewer, matter identifier, and preservation location on every completed set. Preserve observations separately from inferences. Use evidence identifiers consistently across all workpapers.

Classification key

Evidentiary classifications

1. **Known-established:** reliable evidence establishes the proposition and materially plausible alternatives have been addressed.
2. **Known-contradicted:** reliable evidence establishes that the proposed transition did not occur as alleged.
3. **Assumed:** used for a limited analytical purpose but not established; the assumption and effect are explicit.
4. **Undetermined-pending:** potentially resolvable evidence remains to be obtained or tested.
5. **Undetermined-exhausted, absent:** the necessary evidence was not created, retained, or is established not to exist.
6. **Undetermined-exhausted, inaccessible:** evidence may exist, but reasonable acquisition avenues within authority and scope are exhausted.
7. **Undetermined-exhausted, existence unknown:** reasonable inquiries cannot establish whether the evidence exists.

Scope dispositions

8. **Not in issue:** the dimension belongs to the model, but no material fact, context, or competing explanation places it in dispute.
9. **Not applicable:** the dimension does not logically apply to the scoped attribution claim.

These are not evidentiary conclusions and do not mean known-established.

Canonical attribution dimensions

Use these identifiers and names throughout the workpapers:

ID	Canonical dimension	Discontinuity classification
AD-01	External action	Action (7.1)

ID	Canonical dimension	Discontinuity classification
AD-02	Technical actor and execution	Execution (7.2)
AD-03	Credential or account	Identity (7.3)
AD-04	Human identity	Identity (7.3)
AD-05	Natural-person control	Control (7.4)
AD-06	Authorship of the material interaction	Interaction-authorship (7.5)
AD-07	Instruction origin	Instruction-origin (7.6)
AD-08	Knowing authorization	Authorization (7.7)
AD-09	Voluntariness	Voluntariness (7.8)
AD-10	Scope	Scope (7.9)
AD-11	Knowledge	Knowledge (7.10)
AD-12	Intent	Intent (7.11)
AD-13	Delegation	Delegation (7.12)
AD-14	Causal contribution	Causal-contribution (7.17)

Individual and organizational investigative attribution are synthesized conclusions. Responsibility referral is a downstream output.

Discontinuity classification

Record:

10. the affected attribution dimension;
11. the failure condition or conditions;
12. the graph location; and
13. whether the discontinuity is primary, co-primary, contributing, causal, or downstream.

Matter control sheet

Field	Entry
Matter identifier	
Methodology version	Version 0.5
Practitioner	
Independent reviewer	
Date opened	
Application level	Triage / Standard / Enhanced
Mode	Forensic / Assurance
Evidence-preservation location	
Workpaper access classification	

WP-01 Attribution Question and Scope Record

Field	Entry
Consequential action	
Proposed human or organizational attribution	
Attribution dimensions in issue	
Period and material time	
Unit of analysis and aggregation basis	
Action, run, session, episode, incident, campaign, and population counts	
Attempted, executed, completed, effective, harmful, mitigated, and recovered states in scope	
Systems and trust domains	
Included questions	
Excluded questions	
Mode: forensic or assurance	
Application level and justification	
Dimensions not in issue	
Dimensions not applicable	
Individual path, organizational path, or both	
Technical, operator, cluster, organizational, or sponsorship paths in scope	

WP-02 Attribution Graph and Transition Register

Transition ID	From	To	Proposition	Material to which conclusion	Expected evidence	Trust-boundary crossing	Topology or contribution role
T-01							

Attach the graph or identify its preservation location:

Field	Entry
Graph location	
Topology	Hierarchical / peer-to-peer / mesh / hybrid / undetermined
Material forks, merges, or feedback loops	
Stored-state crossings	
Trust transformations	
Later constraints, revocations, or narrowed approvals	
Component, definition, or configuration state changes	

Field	Entry
Decision-basis state and effective-target context	
Shared cross-run state or causal influence	
Human decision gates	

Operative-state record

State ID	Instruction, constraint, component, definition, decision basis, target context, shared state, or gate	Authority or origin	Provenance, version, freshness, and fitness	Delivery or deployment evidence	Effective period	Components, runs, or actions affected	Continuity limit
ST-01							

Approval and effective-target comparison

Complete where approval, confirmation, silence, continued operation, or non-intervention is material.

Approval event	Positive grant and authority	Person or mechanism controlling response	Action, target, decision basis, parameters, risk, and consequence presented	Effective action and target	Material delta	Authorization and knowledge effect
AP-01						

WP-03 Proposition-Specific Evidence and Independence Record

Complete one row for every proposition–stream relationship. The same record may be independent for one proposition and dependent or irrelevant for another.

Proposition	Stream ID	Evidence and independent face fact	Integrity and completeness	Capture mechanism	Dependencies or inherited fields	Control plane or custodian	Common failure path	Evidentiary contribution and limit
	S-01							

Derivative-source record

Complete when an incident registry, postmortem, technique catalog, taxonomy, research reconstruction, or later report repeats or transforms another source.

Derivative artifact	Version and retrieval date	Originating source or evidence	Added structure or analysis	Inherited propositions	Independent observation, if any	Stream-count treatment

Single-stream known-established exception

Complete only where a ceiling-setting proposition is proposed as known-established without two proposition-relevant streams.

Proposition	Reason second stream is unavailable or unnecessary	Direct observation or inherited assertion	Alteration capability	Expected confirming or contradictory records	Alternatives tested	Residual failure risk	Why classification is not lowered	Second-practitioner concurrence, if required

WP-04 Observation, Inference, and Warrant Log

Entry	Observation	Proposed inference	Warrant required	Support for warrant	Assumption or limit
OI-01					

For testimonial evidence, decompose admissions, denials, recollection failures, and explanations into separate propositions.

Statement ID	Statement made	Proposition asserted	Speaker's interest	Contemporaneity and specificity	Consistency	Independent corroboration	Evidentiary treatment
TS-01							

For subject-system and external analytic representations, establish the statement separately from the proposition it asserts.

Representation ID	Source and statement	Proposition asserted	Direct observation, inherited assertion, or analysis	Independent support or contradiction	Evidentiary treatment and limit
SR-01					

For model reasoning, plans, summaries, scratchpads, or explanations, record narrow proof and prohibited inferences.

Reasoning record	Architecture and capture point	Integrity, completeness, transformation, or redaction	Context and tool-call correlation	Inference proposed	Permitted proof	Prohibited inference
RR-01						

WP-05 Competing-Origin and Intervention Matrix

An alternative must pass both the plausibility and materiality gates.

Transition ID	Hypothesis	Plausibility anchor	Material effect if accepted	Originating source	Immediate influencing source	Content or trust transformation	Supporting evidence	Contrary or missing evidence	Status
T-01									

WP-06 Transition Adjudication Matrix

Transition ID	Proposition	Evidence relied upon	Proposition-specific convergence	Zemi classification or scope disposition	Exhaustion basis, if applicable	Reason	Effect on attribution
T-01							

Exhaustion record

Complete before assigning an exhausted status.

Proposition	Evidence sought	Systems and custodians checked	Material-time and scope coverage	Acquisition steps and authority	Provider or witness responses	Why further inquiry is unavailable or disproportionate	Exhausted state selected

WP-07 Discontinuity Schedule

HAAD ID	Canonical dimension: AD ID and name	Failure condition	Role: primary, co-primary, contributing, causal, or downstream	Graph location	Resolution status	Remaining alternatives	Conclusion ceiling	Required future evidence
H-01								

Primary discontinuity:

Field	Entry
Scoped attribution proposition	
Earliest independently ceiling-setting transition on that causal path	
Why downstream evidence cannot cure it	
Co-primary transition, if any	

WP-08 Bounded Attribution Conclusion

Conclusion layer	Finding	Classification or scope disposition	Attribution boundary and basis
Attempted action			
External action			
Successful completion			
Resulting effect			
Demonstrated harm			
Mitigation and recovery			

Conclusion layer	Finding	Classification or scope disposition	Attribution boundary and basis
Technical actor and execution			
Credential or account			
Human identity			
Natural-person control			
Authorship of the material interaction			
Instruction origin			
Knowing authorization			
Voluntariness			
Scope			
Knowledge			
Intent			
Delegation			
Causal contribution			
Individual investigative attribution			
Organizational investigative attribution			
Responsibility referral			

Required bounded conclusion:

Field	Entry
Nearest established actor or state	
Transition at which attribution stops	
Weakest material transition	
Conclusion that would exceed the evidence	
No demonstrated discontinuity within scope, if applicable	
Express limitations	
Unit of analysis and prohibited generalization	
Cross-run dependence and permitted replication claim	

WP-09 Assurance and Readiness Return

Record prospective requirements separately from retrospective findings. A recommended future control does not establish that the historical transition occurred.

Discontinuity	Evidence that was absent or unreliable	Control or instrumentation requirement	Owner	Priority	Reassessment trigger
H-01					

WP-10 Organizational Attribution Path

Complete for Enhanced application and whenever organizational attribution is in scope.

Transition ID	Organizational actor or function	Role: governing, risk, selection, development, operation, identity, control, monitoring, or response	Decision, authority, notice, control, intervention, or enforcement proposition	Evidence	Classification	Effect on organizational attribution	Readiness implication
OT-01							

Where a threat cluster, parent organization, government, sponsor, or other higher-level actor is proposed, complete one row per layer.

Actor layer	Proposed actor	Linking proposition	Accessible evidence	Source-attributed assessment, if any	Independent corroboration	Classification and ceiling
Technical executor						
Account or infrastructure operator						
Natural person or team						
Analytic cluster						
Organization						
Sponsor or state						

WP-11 MCP-Mediated Action Profile

Complete when an MCP client, server, tool, task, or downstream service is material. Do not collapse the chain into a single “tool use” event.

A. Component and definition record

Component	Product, version, and instance	Operator or trust domain	Effective configuration or definition	Acquisition source	Integrity and completeness limit
Host application					
Model or agent					
MCP client					
MCP server					
Tool					
Downstream service					
Package or deployed handler					

B. Protocol transition register

Transition	Proposition	Request, response, or state record	Correlation identifier and time	Authorization, scope, or consent evidence	Parameter or trust transformation	Downstream corroboration	Classification and limit
Natural person → host interaction							
Host → model or agent							
Model or agent → MCP client							
MCP client → MCP server							
MCP server → tool							
Tool → downstream service							
Downstream service → consequential action							

Append one row for every additional downstream action produced by the same invocation.

Authorized or represented action	Arguments at client	Arguments at server	Handler or implementation delta	Effective downstream request	Resulting action	Authorization and scope effect

C. Preservation checklist

Item	Preserved	Evidence ID or reason unavailable
Effective tool catalogue, description, schema, annotations, and refresh state	Yes / No / N/A	
Package, publisher, repository, vendor-authorization, handler-code, deployment, hash, signature, dependency, and update records	Yes / No / N/A	
User and system instructions and relevant context	Yes / No / N/A	
Model-generated tool call and argument transformations	Yes / No / N/A	
Protocol requests, responses, IDs, timestamps, retries, errors, and cancellations	Yes / No / N/A	
Authorization subject, audience, scopes, elevation, revocation, and consent records	Yes / No / N/A	
Confirmation or elicitation wording, material parameters shown, server identity, and response	Yes / No / N/A	
Positive authorization source, authority, effective target, decision basis, risk, and consequence presented	Yes / No / N/A	
Task or state handles, resumptions, and identity binding	Yes / No / N/A	
Trace context and downstream correlation	Yes / No / N/A	
Downstream action or transaction record	Yes / No / N/A	
Separate correlation for every downstream action or secondary consequence	Yes / No / N/A	
Logging coverage, clock synchronization, retention, and alteration capability	Yes / No / N/A	

D. Bounded MCP findings

Proposition	Finding and basis	HAAD dimensions	Classification	Attribution stopping point
Authenticated client				
Tool invocation				
Interaction authorship				
Particular-action authorization				
Tool-definition integrity				
Tool-implementation integrity				
Authorized-argument to downstream-request delta				
Downstream consequence and causal contribution				

WP-12 Campaign, Activity-Share, and Decision-Gate Profile

Complete when a conclusion aggregates several actions, targets, sessions, or agent instances, or relies on a claim about the proportion of work performed by AI.

A. Campaign and aggregation record

Field	Entry
Campaign or population asserted	
Unit of analysis	
Inclusion and exclusion criteria	
Total and examined episodes or targets	
Successful, failed, and incomplete episodes	
Evidence coverage and sampling basis	
Common and divergent paths	
Permitted level of aggregation	

Run or episode	Agent instance and configuration	Shared persistent state or causal influence	Direction of influence	Independent repetition supported?	Aggregation limit

B. Activity-share record

Activity-share claim	Denominator and unit	Measurement method	Source and coverage	Exclusions and uncertainty	Attribution inference permitted	Attribution inference prohibited

C. Strategic-decision-gate record

Decision gate	Proposed human or system actor	Instruction or approval	Evidence of authorship and control	Scope and consequence presented	Classification	Effect on attribution

Quality-control sign-off

Check	Practitioner	Reviewer
External action established independently of the subject agent's assertion		
Subject-system and external analytic representations separated from the propositions asserted		
Technical identity not treated as natural-person identity		
Device or session control not treated as authorship of every material interaction		
Technical permission not treated as substantive authorization		
Capability, authority, and enforcement adjudicated separately		
Later constraints and operative component state established at the material transition		
Authorization distinguished from scope		
Downstream evidence not used to cure an upstream attribution break		
Every stream counted for a named proposition		
Single-stream exception fully documented and concurred with where required		
Testimonial assertions decomposed and corroboration assessed		
Competing alternatives passed both gates		
Affected dimensions distinguished from failure conditions		
Pending and exhausted states supported by the required inquiry record		
DRT not treated as a waiver or transfer of organizational control responsibility		
Individual and organizational paths adjudicated separately		
Technical, operator, cluster, organizational, and sponsorship paths received separate ceilings		
Vendor, package, publisher, implementation, and natural-person identity propositions not collapsed		
Tool description distinguished from implementation behaviour and downstream request state		
Multiple downstream actions adjudicated separately		
Activity-share claims documented without allocating intent or authorization		
Human decision gates mapped independently of tactical activity share		
Episode-level findings not generalized beyond established coverage		
Action, run, session, episode, incident, campaign, and population units not collapsed		
Cross-run independence established before repetitions counted as independent		

Check	Practitioner	Reviewer
Approval, silence, continued operation, or non-intervention not treated as knowing authorization without presentation-to-effect comparison		
Effective target resolved through command, environment, state, credentials, account, downstream request, and resources		
Decision-basis state tested for provenance, version, freshness, completeness, and fitness		
Recorded reasoning not treated as transparent computation or proof of awareness, belief, deception, or intent		
Attempt, execution, completion, effect, harm, mitigation, and recovery separated		
Provider, model, configuration, scaffold, instance, session, run, tool process, account, and downstream executor identities not collapsed		
Organizational functions separated where their authority, knowledge, control, or contribution differs		
Registries, catalogs, taxonomies, mappings, and later reports traced to originating sources		
Report states the exact attribution boundary or no demonstrated discontinuity within scope		

Sign-off field	Entry
Practitioner name and date	
Reviewer name and date	
Unresolved disagreement	
Required escalation or revision	